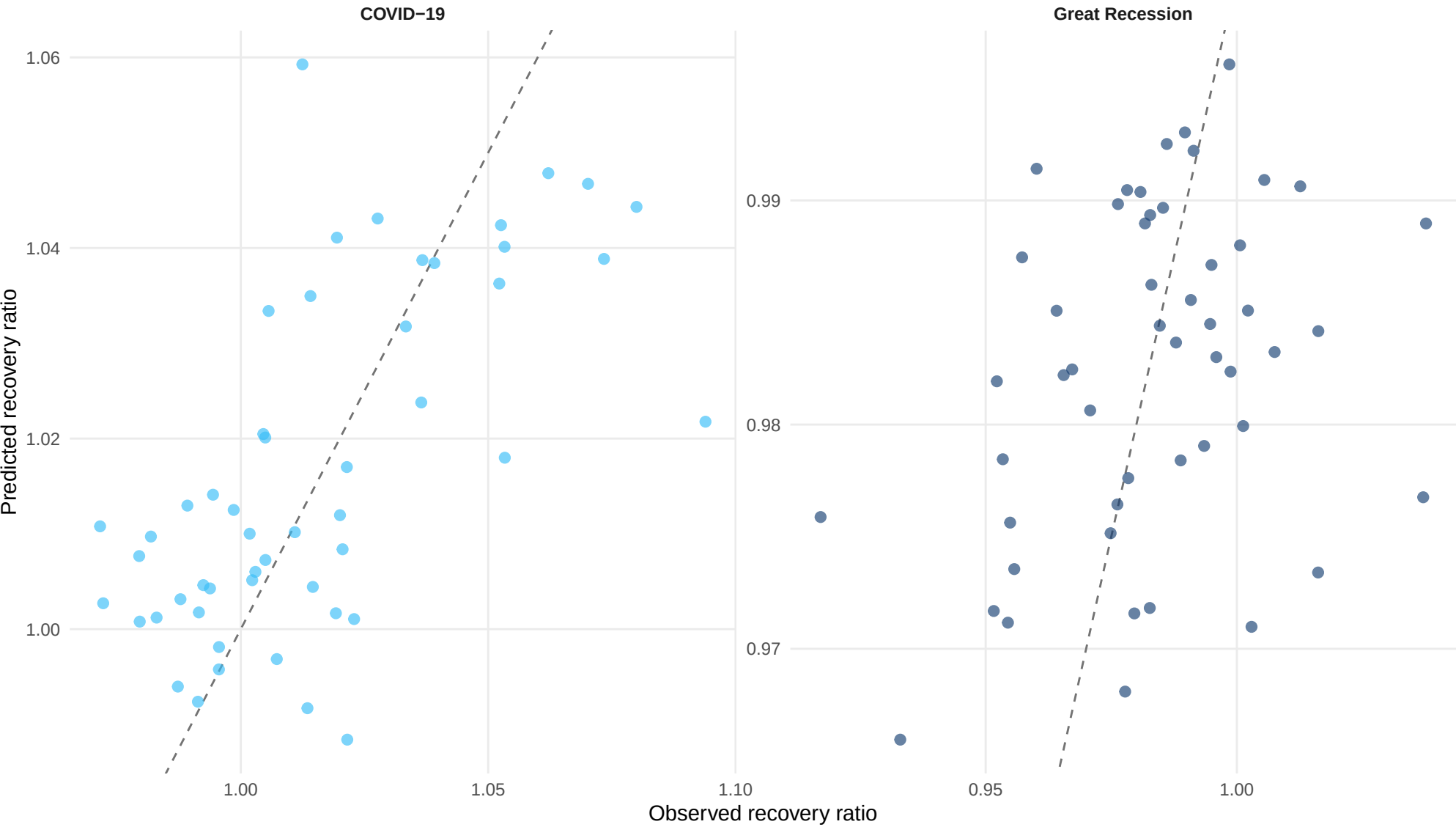


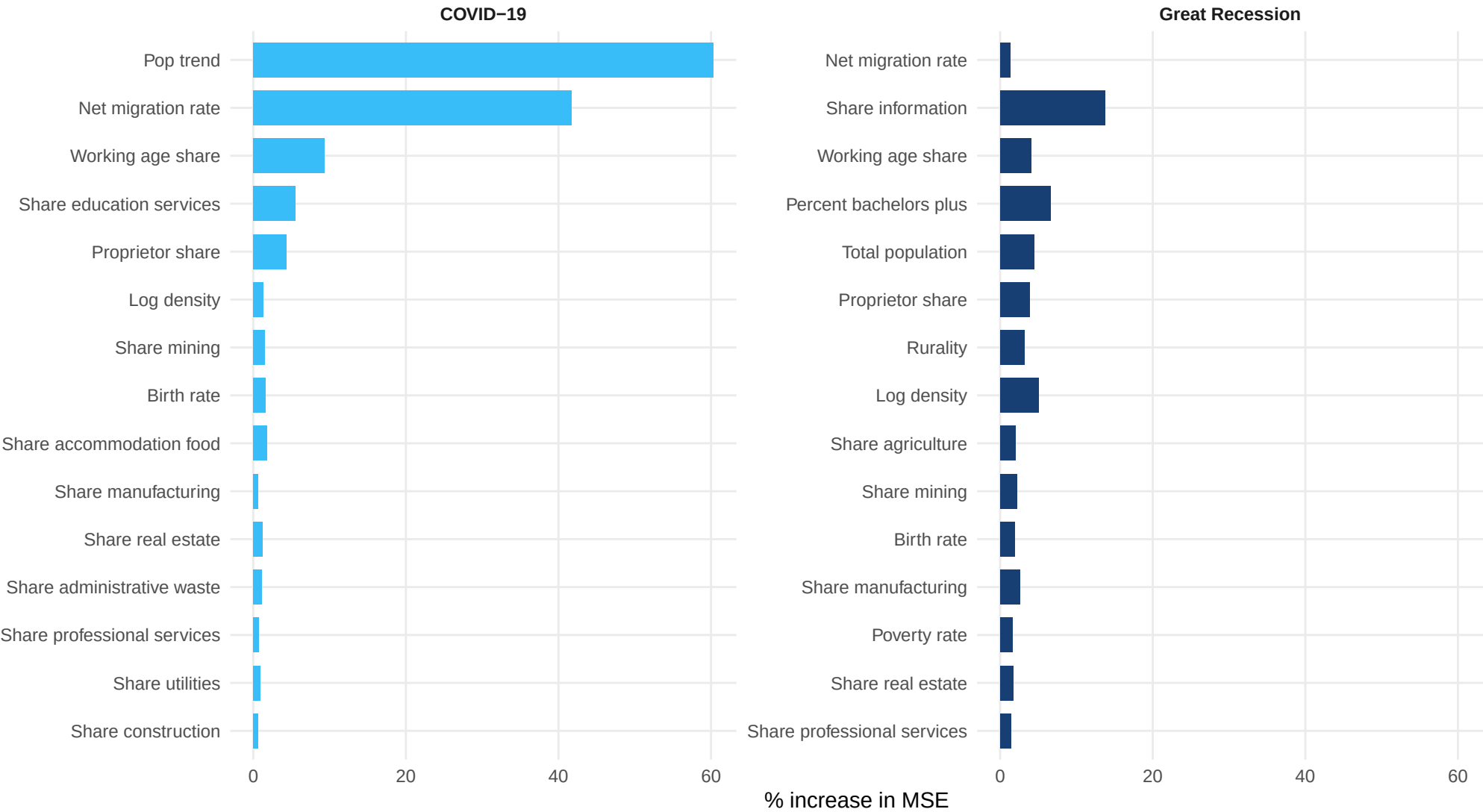
State random forest: observed vs predicted

Out-of-bag estimates for 51 states/DC



State variable importance

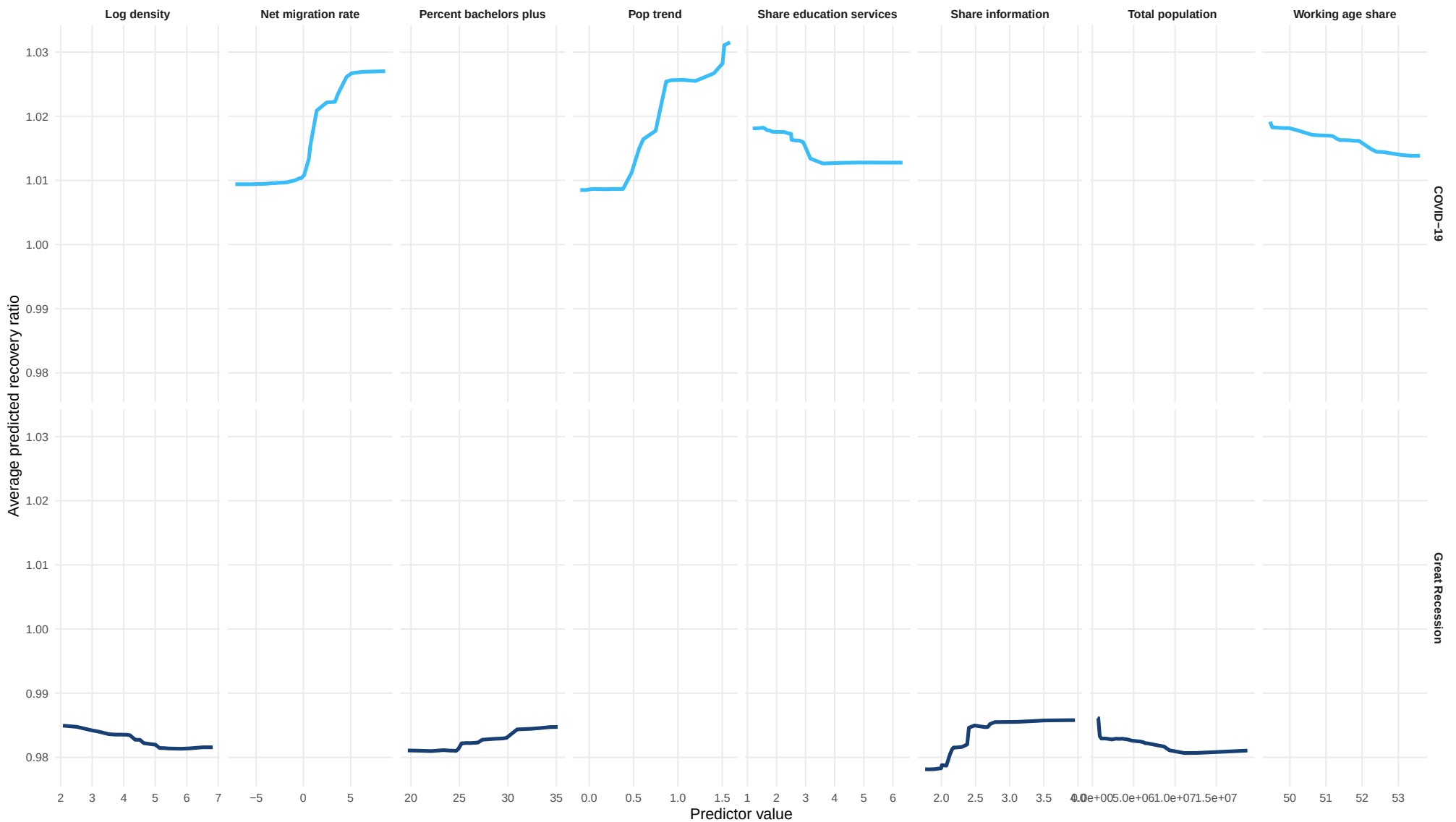
% increase in MSE when predictor is shuffled



Importance indicates predictive reliance, not causality.

State response profiles

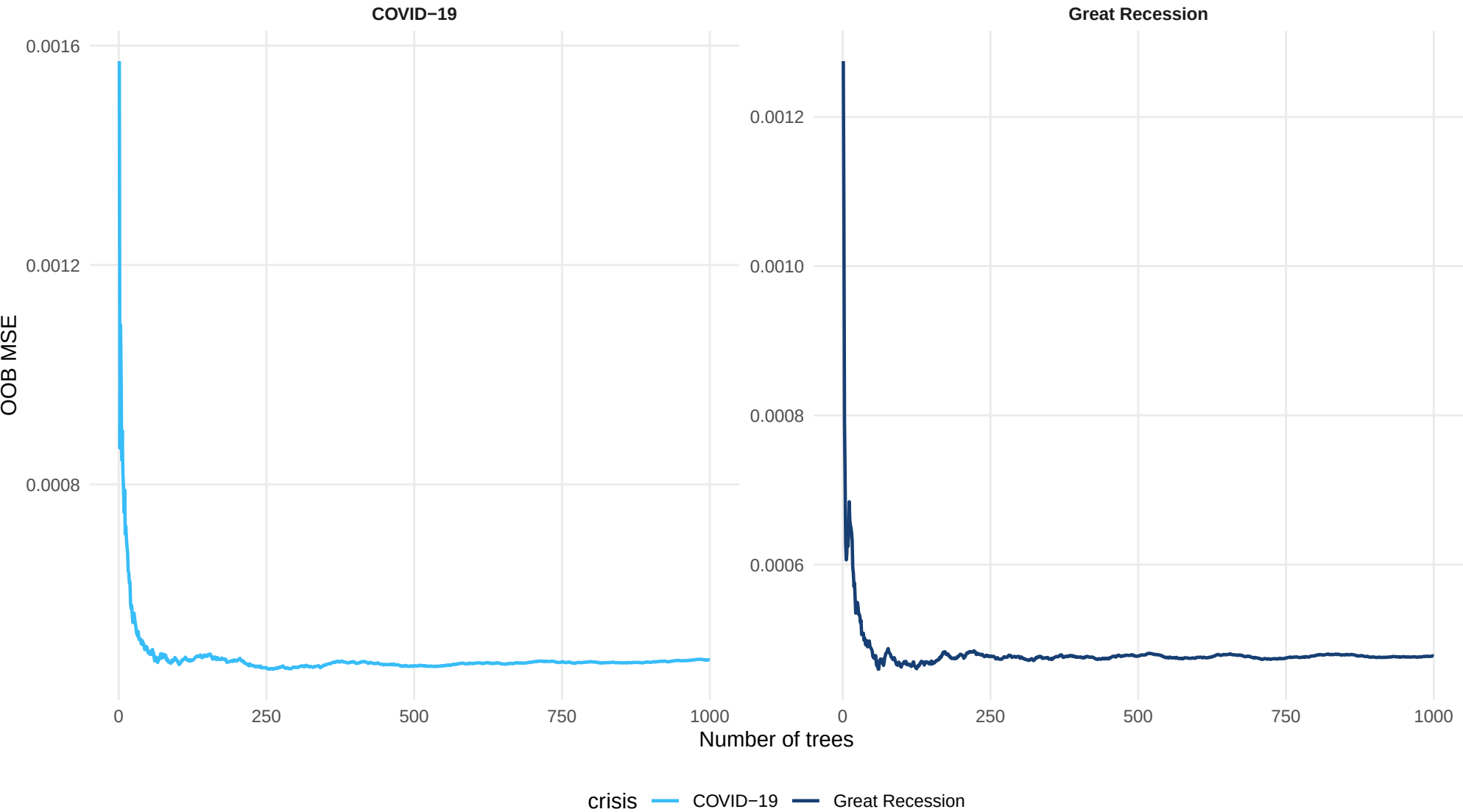
Marginal prediction curves for top numeric predictors



Curves describe model behaviour, not causation.

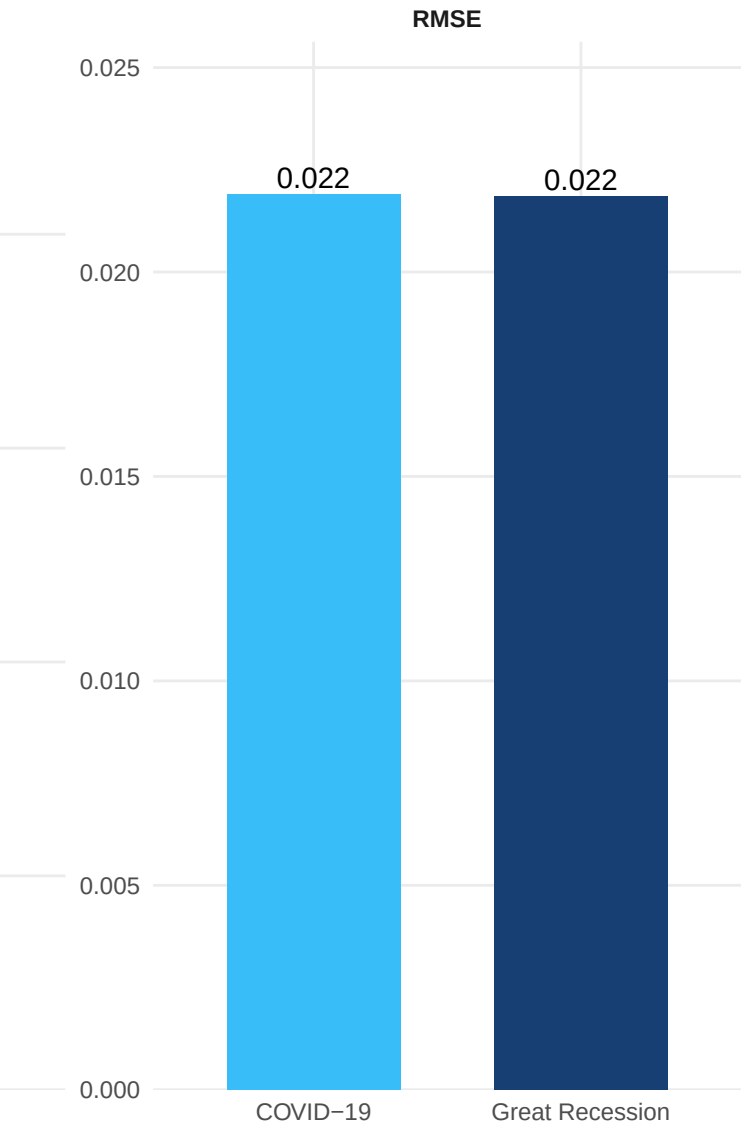
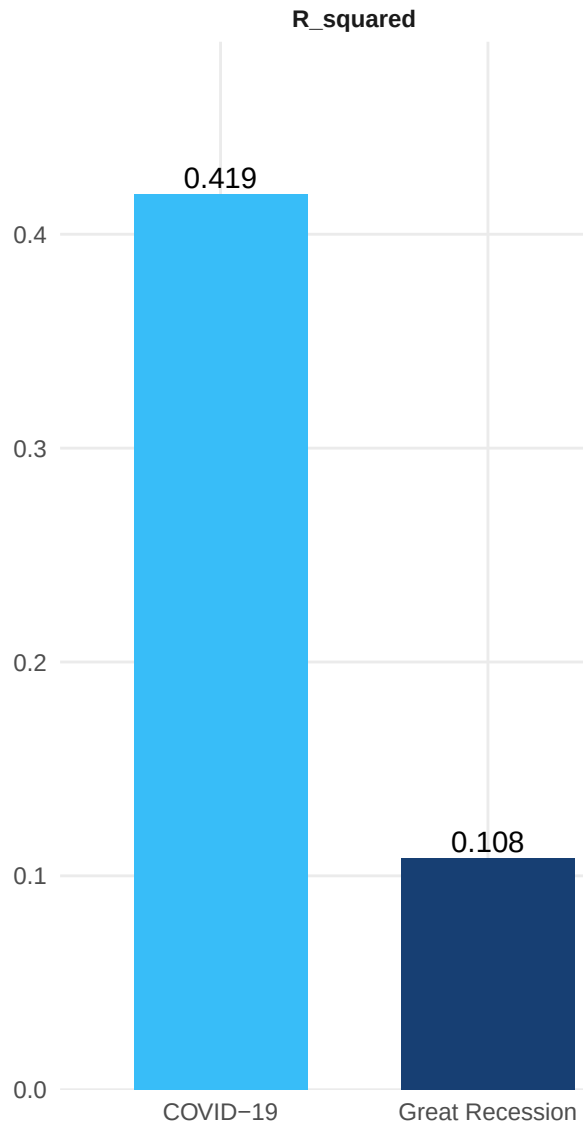
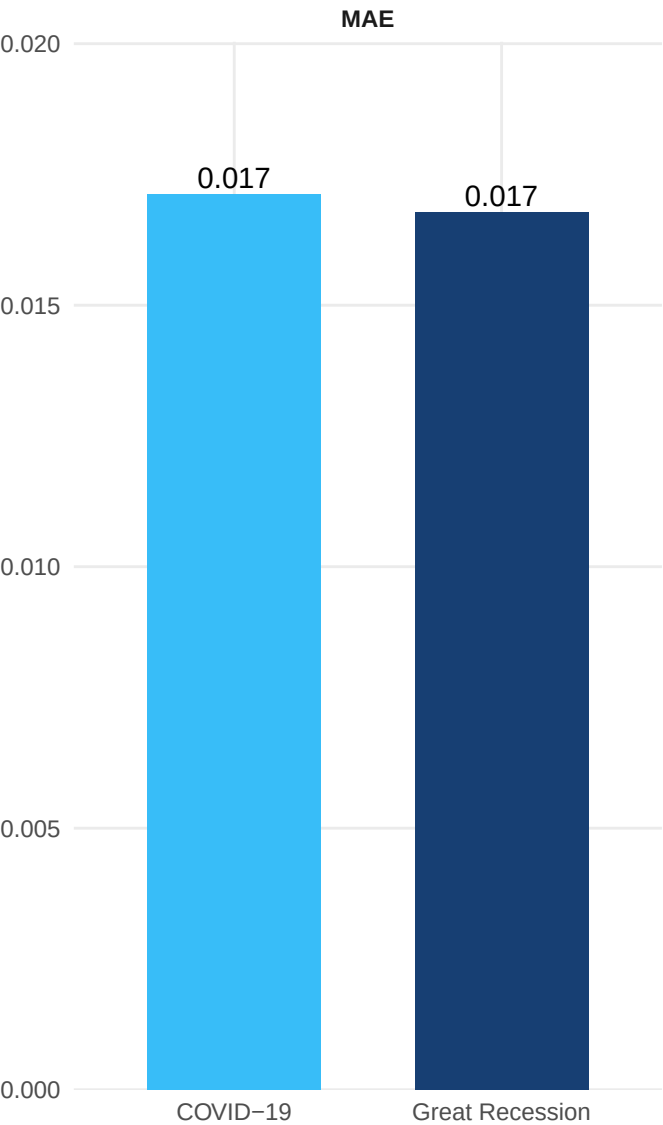
State convergence

OOB MSE stabilises as trees are added



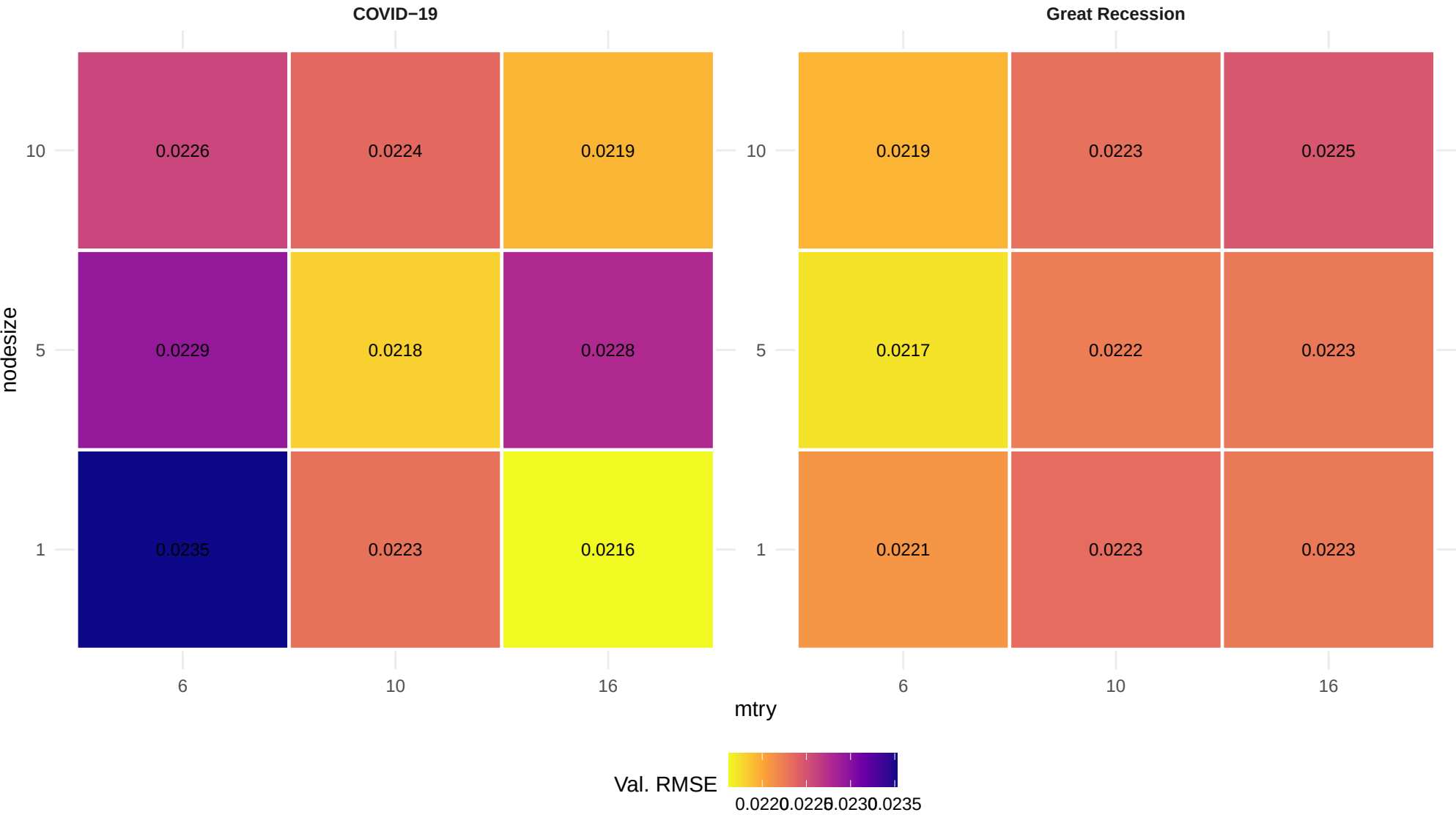
State performance

Lower RMSE/MAE better; higher R² better



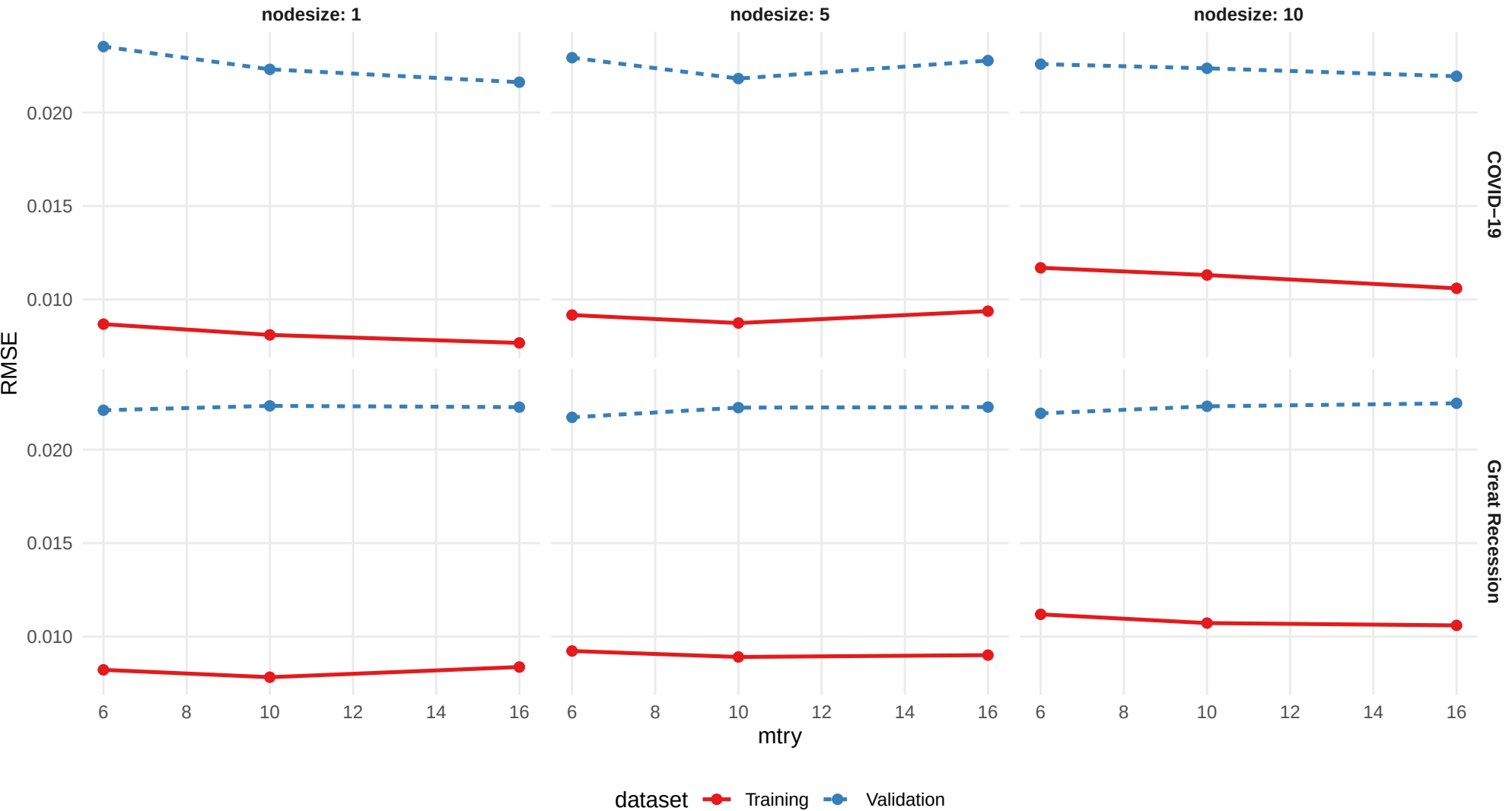
State hyperparameter tuning (val. RMSE)

Validation (OOB) RMSE per mtry / nodesize



State train/validation loss vs hyperparameters

Lines show RMSE for each nodesize across mtry



Validation = OOB error

State Q-Q plot of residuals

Residual normality check

